Davide **Belli**

STAFF ML RESEARCH ENGINEER

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Work Experience

Qualcomm AI Research
Amsterdam, NL

STAFF RESEARCH ENGINEER — *Nov. 2019 - Present*

- Designed and implemented novel Deep Learning methods for **Perception** (visual and RF domains), **Model Efficiency**, and **Agentic AI** (LLMs):
  - Invented a personalized biometric NN classifier improving accuracy from 90% to 99% AUC on challenging real-world data.
  - Implemented Neural KF and WLS methods for GNSS localisation, reducing positioning error by 50% in noisy urban canyons.
  - Designed a dynamic pruning algorithm for recent LLMs (Phi 3, Llama 3, ...), improving throughput by 40% amd memory footprint by 45%.
  - Introduced structured representations and architectures in AI Agents, reducing model footprint by 30x at the same task accuracy.
- Integrated solutions in **products released to customers** (Samsung, Google) and **internal departments**. Filed 11 patent applications.
- Skills and technologies: Python (PyTorch, Hugging Face, vLLM, NumPy, Matplotlib, Pandas, Hydra), Bash, Git, Docker, Run:ai.

University of Amsterdam
Amsterdam, NL

GRADUATE TEACHING ASSISTANT — *Oct. 2018 - Jun. 2019*

- Graduate Teaching Assistant for the core courses Machine Learning 1, Deep Learning and Information Retrieval in the MSc AI at UvA.
- Held lab sessions, prepared and corrected exams, homework and lab assignments.

Education

University of Amsterdam
Amsterdam, NL

MSC IN ARTIFICIAL INTELLIGENCE — *Sep. 2017 - Sep. 2019*

- Graduated **Cum Laude**, GPA 8.8/10 (A+)
- Main Courses: Machine Learning, Computer Vision, NLP, Deep Learning, Reinforcement Learning, Information Retrieval
- Skills and technologies: Python (NumPy, Matplotlib, PyTorch, scikit-learn), MATLAB, Git, LaTeX

University of Trento
Trento, IT

BSC IN COMPUTER SCIENCE — *Sep. 2014 - Jul. 2017*

- Graduated **Cum Laude**, GPA 110/110 (A+)
- Main Courses: Algorithms and Data Structures, Programming, Calculus, Linear Algebra, Probability and Statistics, Logic, Databases
- Skills and technologies: C++, Java, Android, Node.js, Git, HTML/CSS, XML

Publications

- *When Cloud Agents Meet Device Agents: Lessons from Hybrid Multi-Agent Systems*, **Under Review**.
  C Rainone*, D Belli*, B Major, A. Behboodi.
- *Dynamic Tool Dependency Retrieval for Efficient Function Calling*, **ACL 2026** and **AAAI 2026 IR Frontiers workshop (Oral)**.
  B Patel*, D Belli*, A Jalalirad, M Arnold, A Ermolov, B Major.
- *Neural Augmented Kalman Filters for Road Network assisted GNSS positioning*, **ICML 2025 ML4Wireless workshop**.
  H Gorp*, D Belli*, A Jalalirad, B Major.
- *Efficient LLM Inference using Dynamic Input Pruning and Cache-Aware Masking*, **MLSys 2025**
  M Federici*, D Belli*, M van Baalen, A Jalalirad, A Skliar, B Major, M Nagel P Whatmough.
- *GNSS Positioning using Cost Function Regulated Multilateration and Graph Neural Networks*, **ION GNSS+ 2023**
  A Jalalirad, D Belli, B Major, S Jee, H Shah, W Morrison.
- *Online Adaptive Personalization for Face Anti-spoofing*, **ICIP 2022**.
  D Belli, D Das, B Major and F Porikli.
- *A Personalized Benchmark for Face Anti-spoofing*, **WACV 2022 MAP-A workshop**.
  D Belli, D Das, B Major and F Porikli.
- *Image-Conditioned Graph Generation for Road Network Extraction*, **NeurIPS 2019 GRL workshop**.
  D Belli and T Kipf.
- *Adding Object Detection Skills to Visual Dialogue Agents*, **ECCV 2018 SiVL workshop**.
  G Bani*, D Belli*, G Dagan*, A Geenen*, A Skliar, A Venkatesh, T Baumgartner, E Bruni and R Fernández.
- *Context Encoding Chest X-rays*, **arXiv preprint**.
  D Belli, S Hu, E Sogancioglu and B van Ginneken.
- *Chest X-ray Inpainting with Deep Generative Models*, **arXiv preprint**.
  E Sogancioglu*, S Hu*, D Belli and B van Ginneken.

* = Equal contributor

Honors & Awards

- 2025-26 **Top 12%**: Codeforces contest ranking (Global) *Amsterdam, NL*
- 2022-26 **Top 4%**: LeetCode contest ranking (Global) *Amsterdam, NL*
- 2019 **1st Place**: Amsterdam Programming Contest *Amsterdam, NL*
- 2018 **Top 50%**: ACM-ICPC Nwerc (European Semi-Finals) *Eindhoven, NL*
- 2018 **Top 50%**: BAPC Regionals *Louvain-la-Neuve, BE*
- 2016 **Bronze Medal**: ACM-ICPC Swerc (European Semi-Finals) *Porto, PT*
- 2016-19 **Top 15%**: Google Hashcode *Amsterdam, NL*
- 2008-13 **Top 0.1% (7th out of 9000)**: International Championships of Mathematical Games (Italian Finals) *Milan, IT*

Volunteering

- 2020-26 **Conference Reviewer**: WACV (2022-2023-2024-2025), LoG (2023-2024-2025), ICCV (2021)
- 2020-26 **Workshop Reviewer**: ML on Graphs (GLF @ NeurIPS 2022-2023, GLB @ WWW 2022-2023, GRL+ @ ICML 2020)
- 2019-26 **Mentor at LeadTheFuture**: mentorship organization for selected top students in STEM from Italy.
- 2017-19 **Member and Treasurer at Master Committee**: organizing educational events for master students at UvA.
- 2015-17 **School Tutor**: teaching Maths, Physics and Computer Science to High School and Bachelor students.

Patents

Filed 12 patent applications (3 granted, 9 pending):

- Efficient Hybrid Multi-Agent Systems. *US Patent App. 64/055,449, 2026*.
- Optimizing weighted least square (wls) inputs to improve global navigation satellite systems (gnss) localization. *US Patent 12,607,753, 2026*.
- Personalized biometric anti-spoofing protection using machine learning and enrollment data. *European Patent EP4320606, 2026*.
- Architectures and Representations for Agentic AI. *US Patent App. 19/440,522, 2026*.
- System and Method of Tool Dependency Learning for Function Calling AI Agents. *US Patent App. 19/458,639, 2026*.
- System and Method for Efficient Plan Sampling From Function Calling LLMs. *US Patent App. 19/406,680, 2025*.
- Deep Learning methods for Road Network assisted GNSS positioning. *US Patent App. 19/299,160, 2025*.
- Test-time Scaling for Single-turn Function Calling. *US Patent App. 19/270,192, 2025*.
- Adaptive personalization for anti-spoofing protection in biometric authentication systems. *US Patent 12,314,365, 2025*.
- Geometric representation and temporal modeling for GNSS localization. *US Patent App. 19/040,909, 2025*.
- Cache-aware dynamic module selection. *US Patent App. 18/902,554, 2025*.
- Global navigation satellite systems (gnss) localization with residual grid representation. *US Patent App. 18/630,717, 2025*.